Three-dimensional fluorescence recovery with angle-scanned phase illumination and multi-slice fusion

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Abstract

Conventional volumetric fluorescence imaging acquires images at successive focal planes. Depth-dependent illumination offers an alternative: encode the volume into planar measurements and reconstruct it at a fixed detection focus. We combine angle-scanned phase illumination with multi-slice plug-and-play reconstruction. Varying the input angle at a fixed phase encoder changes the volumetric excitation while the specimen and detection focus remain stationary. Reconstruction couples measurement consistency with a pretrained two-dimensional denoiser applied in three orthogonal slice orientations, recovering a shared volume without task-specific three-dimensional network training. Numerical evaluation with reference volumes of mouse pyramidal neurons with dendritic arbors demonstrates recovery over a $30.72 \times 30.72 \times 10.24\,\mu\text{m}^3$ volume with isotropic $0.64\,\mu$m reconstruction sampling. Dominant dendritic structures and their depth ordering remain recognisable in representative reconstructions at eight-fold measurement compression. The framework supports compressive volumetric fluorescence recovery without axial specimen scanning or detection refocusing.

1 Introduction

Fluorescence microscopy reveals labelled structures within cells and tissues. Their spatial organisation often requires three-dimensional imaging, conventionally obtained by recording successive focal planes. Recovering a volume at a fixed detection focus avoids this axial acquisition, but each planar image mixes fluorescence from different depths. The question is how to encode these overlapping contributions so that reconstruction can separate them.

Depth can be encoded in detection or illumination. Light-field microscopy records spatial and angular information, while DiffuserCam and Miniscope3D assign different depths distinct sensor responses [1–3]. Structured and speckle illumination instead weight the volume with known excitation patterns. Prior work demonstrates multilayer fluorescence recovery, compressive recovery from lateral speckle scans, and joint phase and fluorescence imaging [4–6]. Illumination-based depth recovery is therefore established; the question here concerns a fixed phase encoder varied through its input angle.

DEEP and DEEP^2 combine patterned excitation with computational reconstruction to recover fluorescence images through scattering media. Our work follows this excitation-coding approach but addresses volumetric recovery under a negligible-scattering model [7, 8].

We place a phase encoder away from an intermediate illumination image plane and vary the incident angle to generate depth-dependent excitation. A camera records one planar image per angle while the specimen and detection focus remain fixed. Fine illumination structure modulates the fluorescence before detection, and changing the angle gives locations at different depths different intensity sequences. These illumination signatures allow reconstruction to separate contributions that overlap in individual camera images.

Recovering the volume requires a model that connects fine optical structure to the fluorescence represented by each reconstruction voxel. Simulating directly on the reconstruction grid can mis-

represent the measurements. We retain fine-grid optical simulation and integrate the response over each voxel. An analytical illumination–emission factorisation makes this construction tractable by reusing detector responses instead of repeating volumetric convolutions. The derivation was essential to making the simulation and reconstruction workflow computationally feasible within the available resources.

Multi-slice plug-and-play (PnP) reconstruction then combines measurement consistency with a pretrained two-dimensional denoiser applied in three orthogonal orientations. The slice priors refine a shared volume rather than separate reconstructions, following established PnP and multi-slice fusion principles [9, 10]. This avoids training a dedicated three-dimensional network, an important practical advantage when volumetric training data and computational resources are limited. Reconstruction is training-agnostic in the sense that it reuses the pretrained prior without task-specific network training.

The main contributions of this work are:

1. **Angle-controlled volumetric illumination:** A fixed phase-only encoder generates depth-dependent excitation patterns through changes in illumination angle, enabling volume recovery without axial specimen scanning or detection refocusing.
2. **Voxel-integrated reconstruction modelling:** Fine-grid optical responses are integrated over each reconstruction voxel to reduce discretisation-induced model mismatch. An analytical factorisation makes this construction computationally feasible without repeated volumetric simulations.
3. **Training-agnostic multi-slice recovery:** A shared pretrained two-dimensional denoiser regularises the volume through orthogonal slices, avoiding task-specific three-dimensional network training. Representative reconstructions retain dominant dendritic structures and their depth ordering at eight-fold measurement compression.

We evaluate depth placement and foreground structure in eight prespecified mouse pyramidal neuron subvolumes. Eight-fold compression means one scalar detector measurement per eight reconstructed voxel values. Ablations examine voxel modelling, illumination geometry and slice fusion, while scan-design analysis explains the angular coverage and measurement allocation.

The paper is organised as follows. Section 2 presents the acquisition model and reconstruction method, Section 3 describes the simulation setup, Section 4 presents recovery results and ablations, and Section 5 concludes.

2 Methods

2.1 Angle-scanned phase illumination

A tilted plane wave illuminates a fixed phase plate defocused from an intermediate illumination image plane. The excitation relay maps the resulting field into the specimen. Changing the incident direction changes the volumetric illumination, while the plate, specimen and detection focus remain fixed.

The plate represents a phase-only encoder. A spatial light modulator (SLM) could display the same phase profile, held fixed during angle variation. Learning a profile or a stack of diffractive layers would instead provide a diffractive deep neural network (D2NN) encoder, a future extension [11].

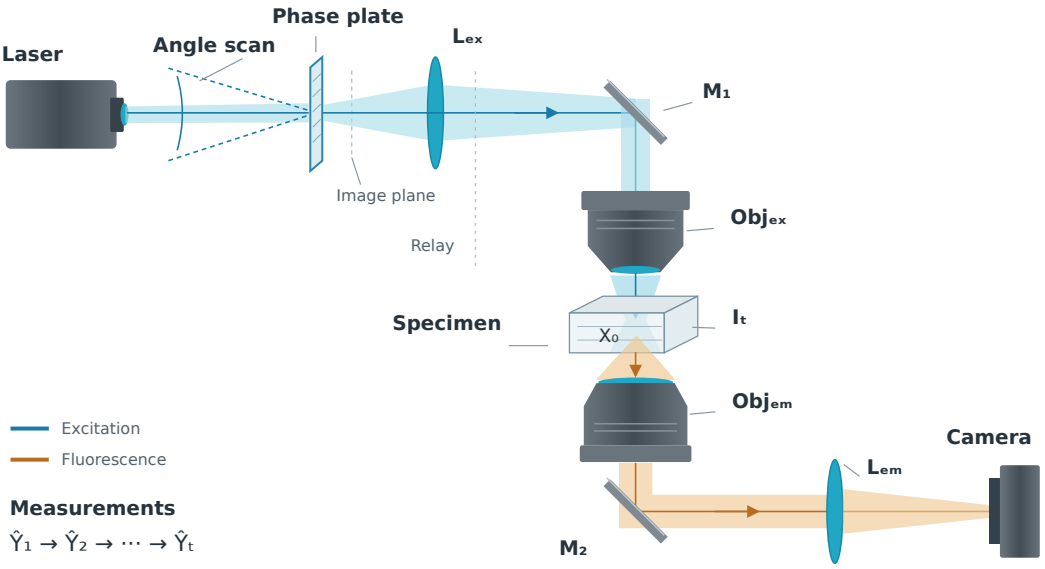


Figure 1: Proposed acquisition. Varying the input angle at a fixed phase encoder generates depth-dependent illumination while the specimen and detection focus remain fixed. Blue: excitation; orange: fluorescence.

Let $\mathbf{r} = (x, y)$, and let t index the illumination direction. The incident field and plate transmission are

$$u_t^{\text{in}}(\mathbf{r}) = a_t e^{\mathrm{i}k_{\text{ex}}\mathbf{s}_t\cdot\mathbf{r}}, \qquad \tau(\mathbf{r}) = e^{\mathrm{i}\phi(\mathbf{r})}. \tag{1}$$

Here a_t is the field amplitude, k_{ex} the excitation wavenumber and $\mathbf{s}_t$ the transverse direction vector. The scan follows directly from plane-wave geometry. With azimuth α_t and elevation θ_t measured from the transverse plane, the propagation direction is $(\cos\theta_t\cos\alpha_t, \cos\theta_t\sin\alpha_t, \sin\theta_t)$. Restricting its phase to the plate plane gives

$$\mathbf{s}_t = \cos\theta_t(\cos\alpha_t, \sin\alpha_t), \qquad \varphi_t(x, y) = k_{\text{ex}}\cos\theta_t(x\cos\alpha_t + y\sin\alpha_t). \tag{1a}$$

Thus the two scan coordinates control the direction and magnitude of the transverse phase gradient. Their sampling determines the sequence of volumetric excitation patterns, rather than changing the detection plane.

Angular-spectrum propagation carries the field through the plate to the relay object plane. High numerical aperture (NA) requires a vectorial description of the transverse and longitudinal field components. The excitation intensity is

$$I_t^{\text{raw}}(\mathbf{r}, z) = \sum_{c\in\{x,y,z\}} |E_{t,c}(\mathbf{r}, z)|^2. \tag{2}$$

We form intensity after field propagation, then average it onto the analysis grid and apply a global scale. The vectorial relay follows the Richards–Wolf formulation and the x-polarised kernel of Marian et al [12, 13].

2.2 Image formation and reconstruction basis

$X_0(\mathbf{r}, z)$ denotes the fluorescence distribution and emPSF the detection intensity point spread function (PSF). Under linear fluorescence,

$$Y_t = \text{emPSF} *_3 (\eta X_0 I_t), \qquad \widehat{Y}_{t,p}^0 = \frac{1}{|D_p|}\int_{D_p} Y_t(\mathbf{r}, z_{\text{focal}})\,\mathrm{d}\mathbf{r}. \tag{3}$$

Here $*_3$ denotes three-dimensional convolution and D_p a detector-pixel region. Fluorescence adds incoherently, and the camera samples the same focal plane for every pattern. The fixed factor η is absorbed into the numerical scale. Each pixel value is one scalar measurement [7, 8].

The multiplication $X_0 I_t$ explains how fine illumination structure contributes to recovery. It mixes specimen and illumination spatial frequencies before fluorescence is blurred and averaged at the detector, allowing some otherwise unresolved specimen structure to influence measurable intensity variations. The detector need not resolve the illumination pattern itself. As the incident angle changes, locations at different depths experience different illumination sequences. Reconstruction uses these depth-dependent signatures to separate contributions that overlap in individual camera images. Depth separation therefore depends on how the illumination varies across the volume and scan, not on fine spatial structure alone.

Representing this modulation accurately requires different sampling grids for optical simulation and reconstruction. We retain fine-grid optical responses rather than simulate directly at the reconstruction spacing. Let B denote the fine-grid measurement process, and let each reconstruction voxel represent constant fluorescence over a fine-grid patch. With patch replication U and patch averaging R, the operator and reference are

$$\mathsf{A} = \mathsf{BU}, \qquad \mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f, \qquad \mathbf{y} = \mathsf{B}\mathbf{x}_f + \boldsymbol{\varepsilon}. \tag{4}$$

Thus each operator column integrates the response over a voxel patch. Measurements are generated from the finer object $\mathbf{x}_f$, retaining within-patch variation that the reconstruction cannot represent. The remaining representation residual is $\boldsymbol{\delta}_{\mathrm{disc}} = \mathsf{B}(\mathbf{x}_f - \mathsf{UR}\mathbf{x}_f)$.

2.3 Tractable construction of the voxel-integrated operator

Voxel integration would otherwise require a volumetric simulation for every operator column and illumination pattern. We make the construction tractable by factorising each response into an illumination value and a reusable emission footprint.

For a unit impulse δ_i at fine-grid position $\mathbf{r}_i = (x_i, y_i, z_i)$, convolution shifts the emission PSF, giving

$$\begin{aligned}(\mathsf{B}_t\delta_i)_p &= I_t(\mathbf{r}_i)\,\mathsf{P}_p[\mathrm{emPSF}(\,\cdot - \mathbf{r}_i)] \\ &= I_t(\mathbf{r}_i)\,g_p(\mathbf{r}_i).\end{aligned} \tag{5a}$$

Here P_p selects the detection plane and averages over pixel p. The footprint g_p is independent of illumination and is computed once for all patterns.

Summing over the fine-grid patch $\mathcal{P}_j$ of reconstruction voxel j gives

$$\mathsf{A}_{(t,p),j} = \sum_{i \in \mathcal{P}_j} I_t(\mathbf{r}_i)\,g_p(\mathbf{r}_i). \tag{5b}$$

This directly constructs the patch-basis operator $\mathsf{A} = \mathsf{BU}$, with the same sampling and padding as the forward model.

For T patterns, P detector samples per image, N reconstruction voxels and F fine-grid voxels, fast Fourier transform (FFT)-based column simulation and footprint-based assembly cost

$$C_{\mathrm{column}} = O(TNF\log F), \qquad C_{\mathrm{assembly}} = O(TPF). \tag{5c}$$

Both expressions exclude shared illumination calculation; assembly also excludes one-time footprint preparation.

The derivation made the volumetric simulation and reconstruction workflow computationally feasible within the available resources. The final operator build, including preparation and validation, takes 109 s on an RTX 4500 Ada GPU.

2.4 Multi-slice plug-and-play recovery

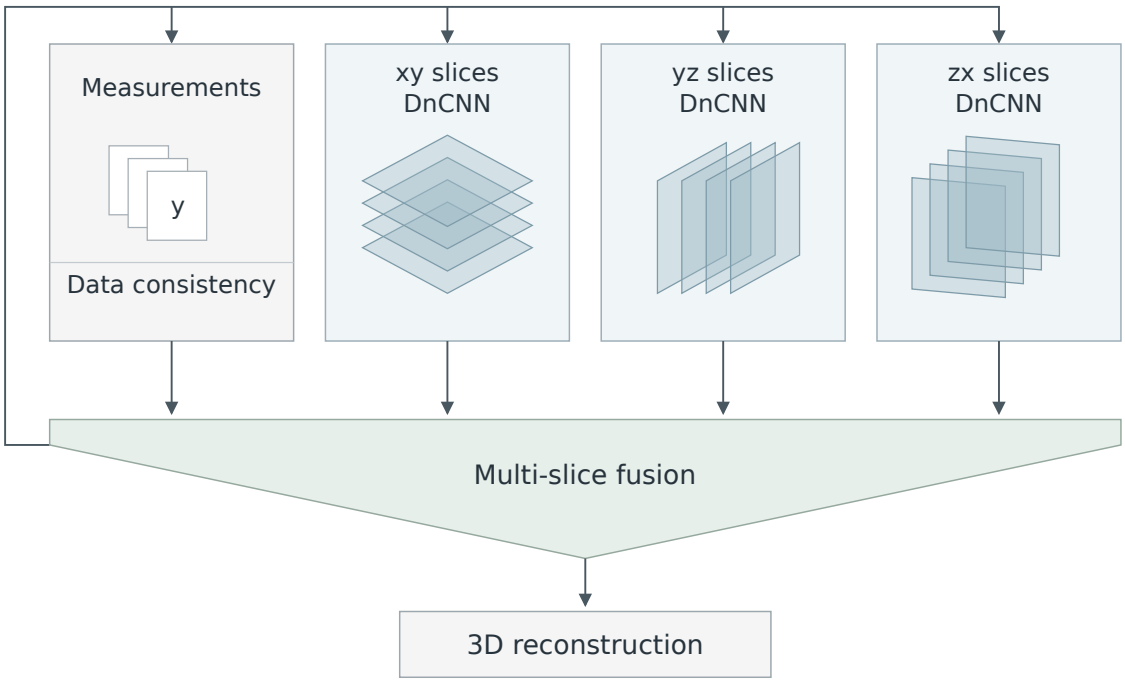


Figure 2: Multi-slice plug-and-play reconstruction. Measurement consistency and orthogonal-slice denoising jointly refine a shared volume.

The planar measurements mix contributions from the volume. We combine measurement consistency with a pretrained two-dimensional denoiser applied to the xy, yz and zx slice sets. The same denoiser weights are held fixed across all orientations and throughout reconstruction: no task-specific denoiser or end-to-end reconstruction network is trained. Each orientation views the same volume, and all branches are coupled through a common update. For orientation o, $\mathcal{D}_o = \mathsf{P}_o^{-1}\mathcal{D}\mathsf{P}_o$, where P_o arranges the slices and $\mathcal{D}$ is the shared denoiser. This follows established PnP and multi-slice fusion principles [9, 10].

We use the alternating direction method of multipliers (ADMM), with one auxiliary volume $\mathbf{v}_j$, dual state $\mathbf{d}_j$ and penalty ρ_j per branch. For $\mathcal{J} = \{xy, yz, zx, \mathrm{sp}\}$, including a sparsity branch, the shared update is

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \left\{ \frac{1}{2}\|\mathsf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \sum_{j\in\mathcal{J}} \frac{\rho_j}{2}\|\mathbf{x} - \mathbf{v}_j^k + \mathbf{d}_j^k\|_2^2 \right\}. \tag{6}$$

The quadratic subproblem is solved approximately by conjugate gradients. The branch and dual updates are

$$\begin{aligned} \mathbf{v}_o^{k+1} &= \mathcal{D}_o(\mathbf{x}^{k+1} + \mathbf{d}_o^k), \qquad o \in \{xy, yz, zx\}, \\ \mathbf{v}_{\mathrm{sp}}^{k+1} &= \mathcal{S}_{\lambda_{\mathrm{sp}}/\rho_{\mathrm{sp}}}(\mathbf{x}^{k+1} + \mathbf{d}_{\mathrm{sp}}^k), \\ \mathbf{d}_j^{k+1} &= \mathbf{d}_j^k + \mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}. \end{aligned} \tag{7}$$

$\mathcal{S}$ is component-wise soft thresholding, with sparsity weight λ_{sp}. The denoiser acts as an implicit prior, and fusion refines one volume throughout the iterations. We initialise with a normalised backprojection and clip the final estimate to $[0, 1]$.

3 Simulation setup

3.1 Optical model and sampling

We simulate a stationary specimen with linear, unsaturated single-photon fluorescence, a homogeneous medium and negligible sample-induced scattering. The excitation objective is modelled with $n = 1$, corresponding to a dry objective. Table 1 lists the numerical configuration. Recovery uses the angular-spectrum implementation.

Table 1: Optical parameters, sampling grids and measurement counts used in the simulations.

Parameter	Value
Excitation / detection wavelength	532 nm, equal-wavelength approximation
Refractive index	$n = 1$
Excitation / detection NA	0.8 / 0.1
Phase-feature pitch / final gap	$0.32\,\mu$m / $6.6\,\mu$m
Illumination directions	16 azimuths × 16 signed transverse directions; 256 camera images
Optical grid	384×384, lateral spacing $0.08\,\mu$m
Analysis grid, (z, x, y)	$32 \times 96 \times 96$, spacing $0.32\,\mu$m
Reconstruction grid	$16 \times 48 \times 48$, spacing $0.64\,\mu$m
Camera	12×12 samples; 8×8 detector averaging

Excitation intensity is averaged over 4×4 lateral optical-grid blocks before analysis. Each reconstruction voxel covers a $2 \times 2 \times 2$ analysis patch. These spacings describe sampling, not optical resolution. Phase values are uniform on $[0, 2\pi)$.

The scan samples $\alpha_i = i\pi/16$, $i = 0, \ldots, 15$, with eight equally spaced elevations in each of $[30°, 60°]$ and $[120°, 150°]$. Opposite signs of $\cos\theta$ provide opposite transverse directions. This covers an annulus in transverse-direction space, $0.5 \leq \|\mathbf{s}\| \leq \sqrt{3}/2$, rather than concentrating measurements along one azimuth. The balanced angular grid follows the preliminary scan analysis described in Section 4.2.

The volume contains $N = 16 \times 48 \times 48 = 36{,}864$ reconstructed voxel values. Each camera image contributes $12 \times 12 = 144$ scalar detector measurements, so $M = 144k$ for k illumination patterns. The 256-direction sequence gives the nominal $M = N$ reference condition. We define the compression factor as $C = N/M$, or equivalently report the measurement-to-voxel ratio $M : N$. The main conditions $C = 1, 4, 8$ use 256, 64 and 32 directions, with ratios $1 : 1$, $1 : 4$ and $1 : 8$. Thus eight-fold compression uses 4,608 scalar measurements to estimate 36,864 voxel values.

For the proposed intermediate-image layout, paraxial scaling with relay demagnification $m_{\text{relay}} = 60$ maps the reference-scale polar tilts of 30°–60° to approximately 8–14 mrad element-side tilts, with $19\,\mu$m plate features and 24 mm defocus.

3.2 Reference objects and noise

Fluorescence reference volumes are drawn from the DEEP^2 dataset of mouse pyramidal neurons with dendritic arbors. We use $32 \times 128 \times 128$ crops from one parent stack, apply min–max normalisation, and area-resample the lateral dimensions to the analysis grid. Each resampled volume is normalised to unit peak intensity and assigned the simulation spacing in Table 1 [8].

Eight subvolumes selected before reconstruction form the quantitative evaluation set. The separate qualitative example was chosen before reconstruction by ranking reference volumes with 1–8% foreground occupancy by projected neurite area multiplied by depth spread, followed by reference-only visual inspection. This selects visible three-dimensional morphology rather than favourable reconstruction quality. Synthetic beads provide a complementary localisation test, with distinct validation and test positions. Measurements are generated from the fine-grid reference objects using the forward model.

The illumination stack is globally normalised to a mean intensity of 30 arbitrary units. Independent zero-mean Gaussian noise is added without clipping. For k illumination patterns, its standard deviation is

$$\sigma_k = \mathrm{mean}(\mathbf{y}^0_{256})\, 10^{-q/10}\sqrt{k/256}, \qquad q = 15. \tag{8}$$

Here $\mathbf{y}^0_{256}$ is the complete noise-free measurement stack; its mean sets the noise scale for every measurement subset. The square-root scaling represents concentrating acquisition effort into fewer images through improved relative measurement precision, rather than keeping per-image noise fixed. Compressed acquisitions use five random illumination subsets at each compression factor. The ablation example, subvolume 0, is evaluated with three noise realisations; the eight-subvolume test set uses one noise realisation per subvolume and the same five subsets.

3.3 Recovery and evaluation

Mouse pyramidal neuron reconstructions use multi-slice PnP with the same pretrained DnCNN-17M denoising convolutional neural network in all orientations; bead localisation tests use the sparsity prior [14]. We select ADMM penalties and the sparsity weight by maximising validation peak signal-to-noise ratio (PSNR) on a separate cortical vascular volume or bead phantom. Parameters are selected for each measurement count and held fixed during testing. For mouse pyramidal neuron recovery at $C = 1, 4, 8$, the denoiser penalties are 5, 1.25 and 1.25 in every orientation; sparsity penalties are 10, 2.5 and 2.5, with weight 0.0125.

All reconstructions use 20 ADMM iterations. Each shared-volume update uses at most 20 conjugate-gradient iterations with tolerance 10^{-5}, providing a common computational budget for comparison.

We measure intensity fidelity with PSNR, foreground structure with the structural similarity index (SSIM), and depth placement with local depth error. PSNR uses the reference intensity range and is interpreted alongside foreground and depth metrics because background can dominate a volume-wide error measure. Foreground SSIM averages a 3D Gaussian-window SSIM map ($\sigma = 1.5$ voxels, data range 1) over reference support above 10% of peak intensity, dilated by two voxels. Local depth error is the reference-intensity-weighted absolute difference between axial centroids of corresponding lateral columns. Columns below 10% of the maximum reference depth-summed intensity are excluded.

For the test set, we average the five illumination subsets within each subvolume, then report the mean and sample standard deviation across the eight subvolumes. Changes are paired with each subvolume's uncompressed reconstruction using matched noise realisations. Separate-example summaries average their noise and illumination-subset repeats.

Bead reconstructions provide a complementary assessment of detection and depth ambiguity. Beads are detected as local maxima after Gaussian smoothing ($\sigma = 0.7$ voxel), with a threshold of 30% of the reconstruction peak. Detections within two voxels laterally and two planes axially of a reference bead are matched, and detection accuracy is summarised by the Jaccard index. The mirror energy ratio compares integrated intensity in 3×3 lateral regions and three-plane windows around the mirror and true depths. Smoothing is used only for detection, not for displayed reconstructions.

4 Results and discussion

4.1 Volume recovery with fewer measurements

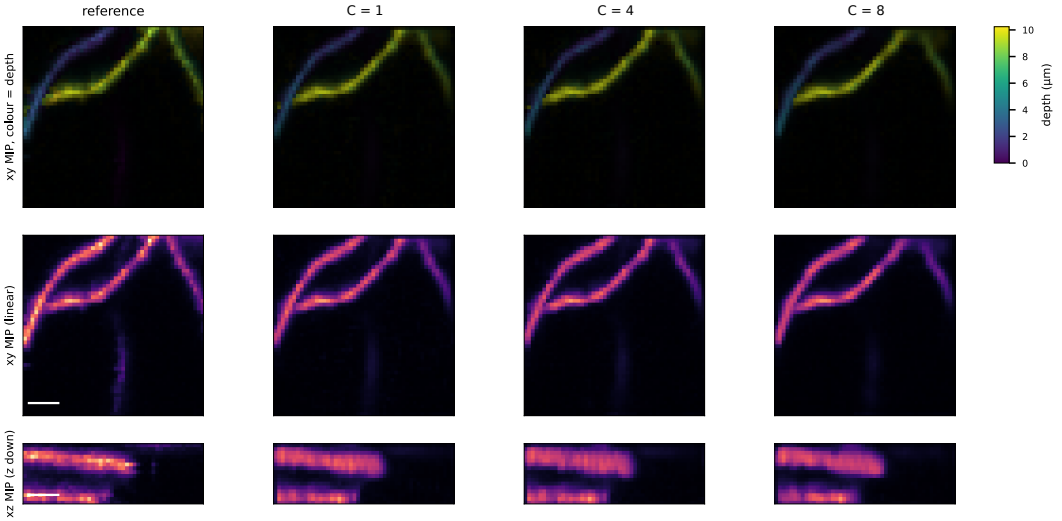


Figure 3: Mouse pyramidal neuron volume recovery at compression factors C=1, 4 and 8 (reference-selected subvolume 384, separate from the test set). Rows: depth-coloured and intensity lateral maximum-intensity projections (MIPs), then xz MIPs. Depth colour indicates the brightest voxel along each column. All panels share intensity limits [0,0.880] and physical aspect ratio; bars: 5 micrometres at the assigned simulation scale.

Angle-scanned phase illumination enables volume recovery from fewer scalar measurements than reconstructed voxel values. In the representative reconstruction shown in Figure 3, dominant dendritic structures and their depth ordering remain recognisable even at eight-fold compression. The depth-oriented views also reveal axial thickening, and a faint vertical neurite is largely lost even without compression. The retained large-scale structure therefore coexists with incomplete recovery of finer features.

Table 2 summarises performance across the eight-subvolume test set. At $C = 8$, mean depth error increases from 0.497 to 0.626 μm, while foreground SSIM decreases from 0.679 to 0.588. The paired depth increase is $0.129 \pm 0.032\,\mu$m, with a PSNR decrease of 0.91 ± 0.45 dB. At $C = 4$, the corresponding changes are smaller: $0.053 \pm 0.016\,\mu$m in depth error and 0.45 ± 0.35 dB in PSNR. Depth error increases in every test subvolume at both compression levels, with four-fold increases ranging from 0.025 to 0.076 μm. These paired comparisons distinguish compression costs from variation between specimen structures.

Table 2: Eight-subvolume test set. Mean and sample SD across subset-averaged subvolumes; paired changes use each subvolume's own uncompressed acquisition.

Compression C	Depth error (μm)	Foreground SSIM	Paired depth change (μm)	Paired SSIM change
1	0.497 ± 0.183	0.679 ± 0.238	0	0
4	0.550 ± 0.170	0.652 ± 0.252	$+0.053 \pm 0.016$	-0.027 ± 0.025
8	0.626 ± 0.207	0.588 ± 0.273	$+0.129 \pm 0.032$	-0.091 ± 0.040

Recovery quality varies across specimen structures. The two dense subvolumes, with approximately 30–34% occupancy, have uncompressed foreground SSIM of 0.44–0.47; the near-empty subvolume has an unstable foreground metric because of its limited support. The separate ablation example, subvolume 0, has depth errors of 0.177, 0.231 and 0.276 μm at $C = 1, 4, 8$. The test set captures variability beyond these separate examples.

Synthetic beads separate detection accuracy from depth placement (Figure 4). The detection Jaccard index remains 1.00 at $C = 1, 4, 8$, but mean depth error increases from 0.183 to 0.282 and 0.544 μm, respectively. The corresponding mirror-to-true-depth energy ratios are 0.011, 0.012 and 0.035. Thus all beads remain detectable under compression, while their depth placement becomes less precise and mirror-depth leakage increases.

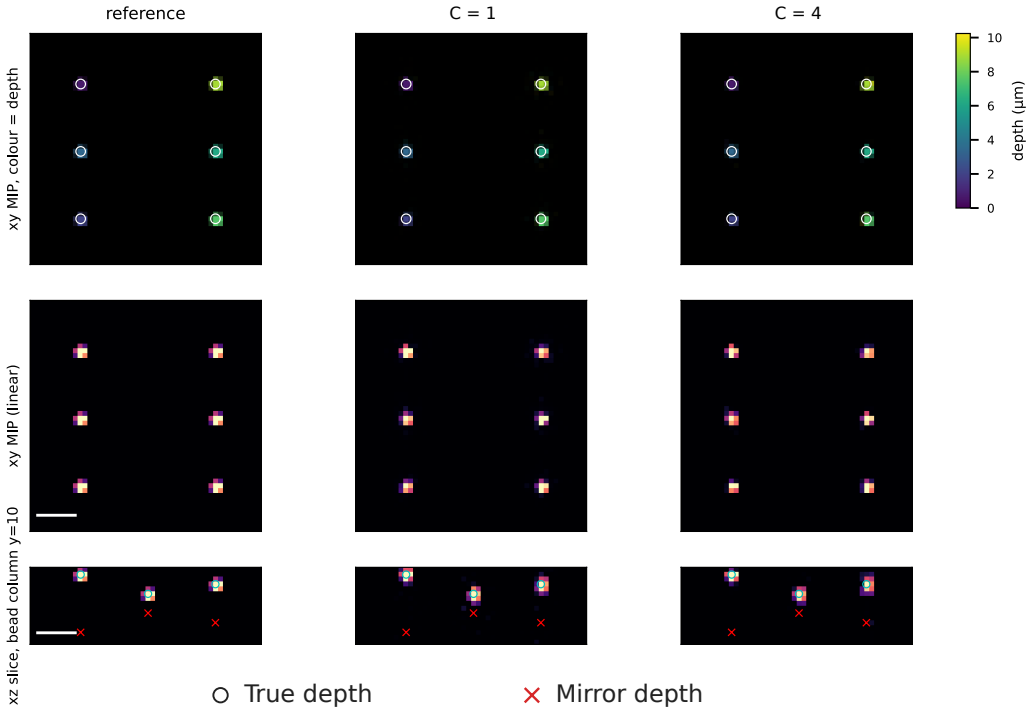


Figure 4: Synthetic bead recovery at C=1 and C=4. Rows: depth-coloured and intensity lateral MIPs, then an xz slice at lateral index 10. Circles mark true positions and crosses mirror positions. All panels share intensity limits [0,1] and physical aspect ratio; bars: 5 micrometres at the assigned simulation scale.

4.2 Ablation studies

The following ablations examine how the measurement model, illumination encoding and reconstruction prior contribute to recovery. We then consider angular coverage and the trade-off at stronger compression.

4.2.1 Model mismatch and voxel integration

We compare voxel-integrated fine-grid modelling with direct optical simulation at the 0.64 μm reconstruction spacing. Both reconstruct identical fine-grid measurements with the same iteration budget and separately validated parameters.

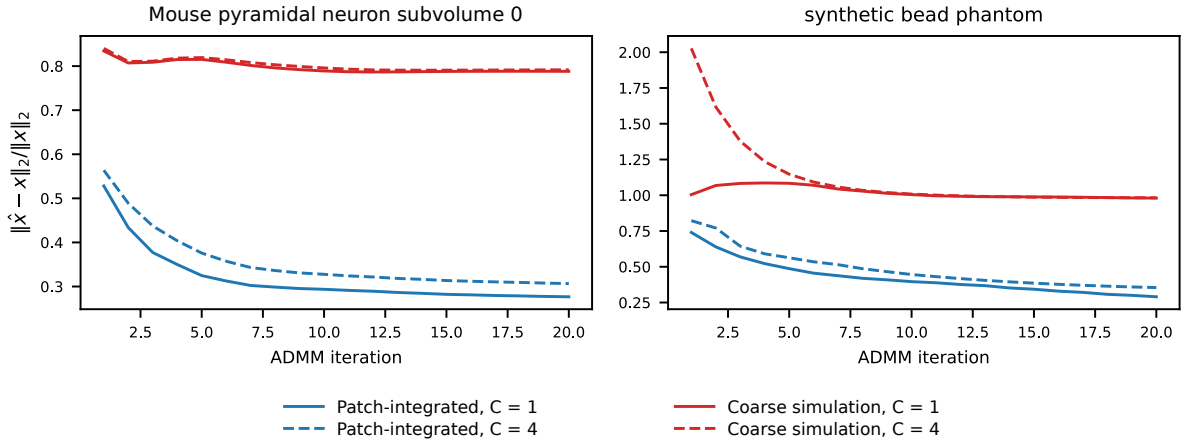


Figure 5: Effect of voxel modelling on reconstruction error. Patch-integrated and coarse-simulation curves average three noise realisations; solid/dashed lines denote C=1/C=4.

For subvolume 0, patch integration reduces the noise-free relative forward discrepancy from 25.7% to 1.07% and improves reconstruction (Table 3, Figure 5). The difference has a physical basis: the high-NA field requires lateral sampling no coarser than $\lambda/(2\mathrm{NA}) \simeq 0.33\,\mu$m, so direct simulation at $0.64\,\mu$m cannot represent its full spatial bandwidth.

Table 3: Voxel-model ablation: mouse pyramidal neuron subvolume 0, uncompressed acquisition, mean over three noise realisations. Depth errors are in micrometres, not voxel planes.

Operator	Depth error (μm)	Foreground SSIM	PSNR (dB)
Patch-integrated	0.177	0.913	37.22
Coarse optical simulation	1.859	0.356	28.12

The advantage persists at $C = 4$, with depth errors of $0.236\,\mu$m for patch integration and $1.867\,\mu$m for coarse simulation. Accurate optical modelling therefore remains important under measurement compression.

4.2.2 In-focus DMD baseline

A digital micromirror device (DMD) generates illumination patterns by amplitude modulation, as in DEEP and DEEP^2. We compare this alternative with the phase encoder to test whether amplitude-pattern variation provides the depth information required for volumetric recovery.

For ideal scalar propagation through an aberration-free symmetric relay, a real-valued focal-plane field U gives conjugate fields at opposite defocus and hence equal excitation intensities:

$$U(\mathbf{r}, -\Delta z) = U(\mathbf{r}, +\Delta z)^*, \qquad I(\mathbf{r}, z_{\mathrm{focus}} - \Delta z) = I(\mathbf{r}, z_{\mathrm{focus}} + \Delta z).$$

A defocused phase encoder supplies a complex field at the relay image plane, removing this symmetry constraint. The vectorial simulations show the same contrast between the two geometries.

To distinguish mirror ambiguity from local depth sensitivity, we use the point-emitter response $\mathbf{a}(z)$ at fixed lateral position. With unknown brightness b and Gaussian noise scale σ, the axial

information and mirror separation are

$$\begin{aligned} \mathcal{I}_z &= \frac{b^2}{\sigma^2}\left(\|\partial_z \mathbf{a}\|_2^2 - \frac{(\mathbf{a}^\top \partial_z \mathbf{a})^2}{\|\mathbf{a}\|_2^2}\right), \quad \sigma_z \geq \mathcal{I}_z^{-1/2}, \\ d' &= \min_c \frac{\|b\mathbf{a}(z) - c\mathbf{a}(z_m)\|_2}{\sigma}, \qquad z_m = 2z_{\text{focus}} - z. \end{aligned} \tag{9}$$

Here c fits the brightness at the mirror depth. Medians cover 144 lateral positions and interior depth planes, with 0.32 μm central differences and the focus excluded from mirror comparisons. The diagnostic uses $b = 1$, $q = 25$; recovery uses $q = 15$.

The DMD and phase plate have similar local axial bounds, 0.02183 and 0.01885 μm, but markedly different mirror separation: $d' = 0.0379$ and 44.76, respectively. Local depth sensitivity therefore does not guarantee separation of mirror-related positions; these bounds are not optical-resolution measurements.

With the same pretrained denoiser and solver and separately validated parameters, uncompressed neuronal reconstruction gives depth errors of 1.775 μm for the DMD and 0.177 μm for the phase plate. The DMD error approaches the empty-volume value of 1.88 μm, showing that the prior cannot replace depth-distinguishing measurements. Bead mirror energy ratios of 1.007 and 0.011 confirm the contrast. This comparison concerns the evaluated in-focus geometry and uses each encoder's relative-noise scale, not matched calibrated photon counts.

4.2.3 Effect of slice fusion

To assess the reconstruction prior, we compare fused and individual slice orientations on uncompressed subvolume 0. Three-orientation fusion reaches 37.22 dB PSNR, compared with 36.26–36.81 dB for individual orientations and 33.99 dB for sparsity alone. The best single orientation, xy, has slightly lower depth error than fusion (0.163 versus 0.177 μm); fusion improves intensity fidelity, not every metric. Removing sparsity from the fused configuration leaves PSNR at 37.22 dB, indicating no additional PSNR benefit at the evaluated sparsity weight.

4.2.4 Angular coverage and measurement allocation

At a fixed pattern count, our preliminary scan-design experiments found lower maximum pairwise output-field correlation with balanced azimuth/elevation allocations than with strongly unbalanced grids. Detector-size experiments also showed that allocating more measurements to illumination diversity could improve recovery. These findings motivate balanced angular coverage and the measurement-count evaluation.

Equation (1a) explains why equal angular increments need not give equal phase-gradient changes. With $\rho = |\cos\theta|$, direction-space separations are $2\rho|\sin(\Delta\alpha/2)|$ for azimuth steps and $|\cos(\theta + \Delta\theta) - \cos\theta|$ for elevation steps. Since $M = Pk$, fewer detector samples per image permit more illumination directions at a fixed measurement count. We fix the detector, evaluate $k = 256, 64, 32$, and draw compressed subsets across the full angular grid.

4.2.5 Higher compression factors

Extending the compression test on subvolume 0 gives foreground SSIM of 0.832 and 0.761 at $C = 16, 32$, compared with 0.913 at $C = 1$. Depth errors rise to 0.354 and 0.443 μm. These exploratory results show the growing trade-off; the main assessment remains $C = 1, 4, 8$ across eight subvolumes.

4.3 Discussion and limitations

The results connect depth encoding, voxel-integrated modelling and multi-slice recovery. The illumination supplies depth-distinguishing measurements, the integrated model represents their dependence on the volume, and slice priors regularise the inverse problem. Representative reconstructions retain dominant dendritic structures at eight-fold compression, while the test set quantifies the fidelity trade-off. Recovery from angular subsets is consistent with redundancy among correlated illumination patterns; independent patterns are not required or assumed.

The acquisition builds on illumination-coded volumetric microscopy. Speckle microscopy already recovers depth from lateral scans, including in nonsaturated fluorescence [5]. Within this family, our study evaluates angle variation at a fixed phase encoder together with voxel-integrated modelling and multi-slice reconstruction. The mismatch ablation explains why retaining fine optical responses and integrating over the reconstruction basis improves recovery, while the analytical factorisation makes that model practical to construct.

This study is numerical. The optical model assumes linear fluorescence in a homogeneous, negligible-scattering medium and equal excitation and detection wavelengths. Its additive Gaussian noise is a relative-noise model rather than calibrated photon noise, and reconstructions use a fixed iteration budget. Sampling intervals, centroid errors and local point-emitter bounds should not be interpreted as optical resolution.

Experimental implementation would require calibrated illumination and detection responses and controlled alignment. A phase-only SLM could implement the encoder, while learned phase profiles may improve recovery at stronger compression. Within PnP, a task-specific three-dimensional denoiser could exploit volumetric structure more directly and potentially support stronger compression when sufficient training data and computational resources are available. These learned extensions remain future directions.

5 Conclusion

We presented a numerical framework for fixed-focus volumetric fluorescence recovery using angle-scanned phase illumination and multi-slice PnP. Fine-grid optical simulation and voxel integration reduce discretisation-induced model mismatch, while an analytical illumination–emission factorisation makes the workflow computationally feasible. Reconstruction reuses a pretrained two-dimensional denoiser without task-specific three-dimensional network training.

Dominant dendritic structures and their depth ordering remain recognisable in representative reconstructions at eight-fold measurement compression. The eight-subvolume test set quantifies the fidelity trade-off, while ablations show the benefits of depth-distinguishing illumination and voxel-integrated modelling. The framework supports volumetric recovery without axial specimen scanning or detection refocusing. Hardware validation, learned phase profiles and task-specific three-dimensional denoisers for potentially stronger compression remain future work.

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